Census Income Classification & Marketing Segmentation

A two-layer targeting design for retail marketing on weighted CPS data

199,523 records · 42 features · CPS 1994–1995

PROJECT STAGES

1	2	3	4	5	6	7
Setup	Profiling	Design	Preprocess	Baseline	Tuning	Segments

8 Package

Executive Summary

This project builds two complementary tools for retail targeting on weighted CPS census data:

0.70	0.96	3
TEST PR-AUC	TEST ROC-AUC	MARKET SEGMENTS

0.25
BALANCED THRESHOLD

The core insight

The final deliverable is not "just a model." It is a two-layer targeting design: the **classifier** tells you *who to prioritize*; the **segmentation** tells you *how to approach them*. Neither alone is the full answer.

Classifier: CatBoost on a reduced feature set (31 features). Tuned with a 27-configuration grid search. Three operating modes (precision-leaning, balanced, recall-leaning) give the business control over the targeting tradeoff.

Segmentation: K-Prototypes clustering on 17 mixed-type features produces 3 interpretable life-stage groups — seniors, youth/dependents, and working-age householders — each suggesting distinct marketing approaches.

1. Problem Framing

A retail client wants to identify high-income individuals for premium product targeting. Census data provides demographic and employment variables but no direct transaction data. Income above $50,000 serves as a **proxy target** — not the ultimate business objective.

> **Income is a proxy**
>
> What the client actually cares about is purchasing power, product affinity, and propensity to respond. The >$50K label is a useful first approximation, but a production system should validate against real business outcomes.

The Targeting Pipeline

The business decision happens at the threshold layer, not at the model output.

2. Data Understanding

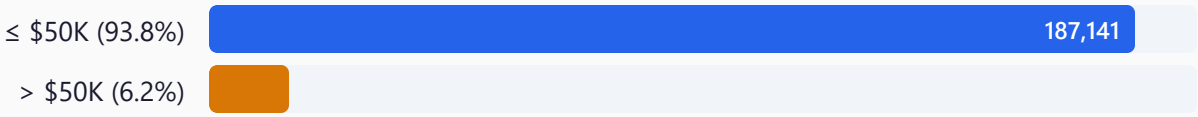
199,523	42	6.2%
RECORDS	COLUMNS	POSITIVE CLASS

94 / 95

SURVEY YEARS

Target Imbalance

The >$50K class represents only 6.2% of records. A naive "predict everyone as ≤$50K" model achieves 93.8% accuracy — illustrating why accuracy is a misleading metric for this task.



12,382 positive-class records out of 199,523 total

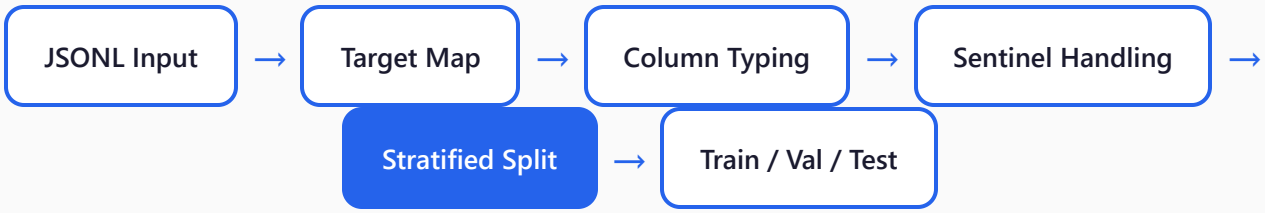
Key Data Characteristics

CHARACTERISTIC	DETAIL
Sentinel values	`Not in universe` appears in 15+ columns (50–99% of rows). Encodes life-stage status — preserved as explicit category, not treated as missing.
Unknown values	? mapped to `Unknown` — ambiguity signal preserved.
Numeric-coded categoricals	5 columns (industry/occupation codes, veterans benefits, year) look numeric but are semantic codes — cast to string to prevent accidental continuous treatment.

Survey weight	CPS population weight (range 38–18,656). Used for evaluation and segment sizing, never as a feature.
Skewed financials	Capital gains, dividends, wage per hour: heavy zero-inflation and right skew. Log1p transforms available for linear models.

3. Preprocessing & Governance

Pipeline Flow



Key Governance Decisions

DECISION	CHOICE	RATIONALE
`weight` column	Excluded from features	Survey weight for population representativeness, not a predictor
`Not in universe`	Preserved as category	Encodes life-stage (children, retirees) — highly predictive
Numeric-coded columns	Cast to string	Prevents accidental mean/scaling on semantic codes
Train/Val/Test split	70/15/15, stratified on target	Preserves 6.2% positive rate across all splits
Encoder fitting	Train-only	Prevents data leakage from validation/test
Review columns (5)	Tested full vs reduced	92–99% sparse; reduced set performed equal or better

Duplicate leakage audit

> Checked before modeling: 1.6% exact duplicates (expected in census microdata), 89 conflicting-target groups (different people with identical demographics). Risk: LOW-MODERATE. No rows removed.

4. Classification Modeling

Stage 5: Baseline Comparison

Eight configurations tested: 2 model families × 2 feature sets × 2 training modes.

CONFIGURATION	PR-AUC	ROC-AUC	F1 @0.5
LR / Full / NoClsBal	0.5978	0.9416	0.4835
LR / Reduced / NoClsBal	0.5976	0.9416	0.4835
CB / Full / NoClsBal	0.6865	0.9519	0.5719
CB / Reduced / NoClsBal	**0.6877**	**0.9517**	**0.5714**
CB / Reduced / ClsBal	0.6735	0.9523	0.4783

CB = CatBoost, LR = Logistic Regression. ClsBal = class-balanced training. Validation metrics shown.

Three key findings

CatBoost outperformed LR by **+0.09 PR-AUC**. Reduced features ≥ Full (sparse columns add noise). No-class-balancing outperformed class-balancing for PR-AUC.

Stage 6: Tuning & Final Model

27-configuration grid search (depth × learning_rate × l2_leaf_reg) with early stopping.

0.7008
TEST PR-AUC

0.9565
TEST ROC-AUC

0.6276
TEST F1 @T=0.25

0.0311

BRIER SCORE

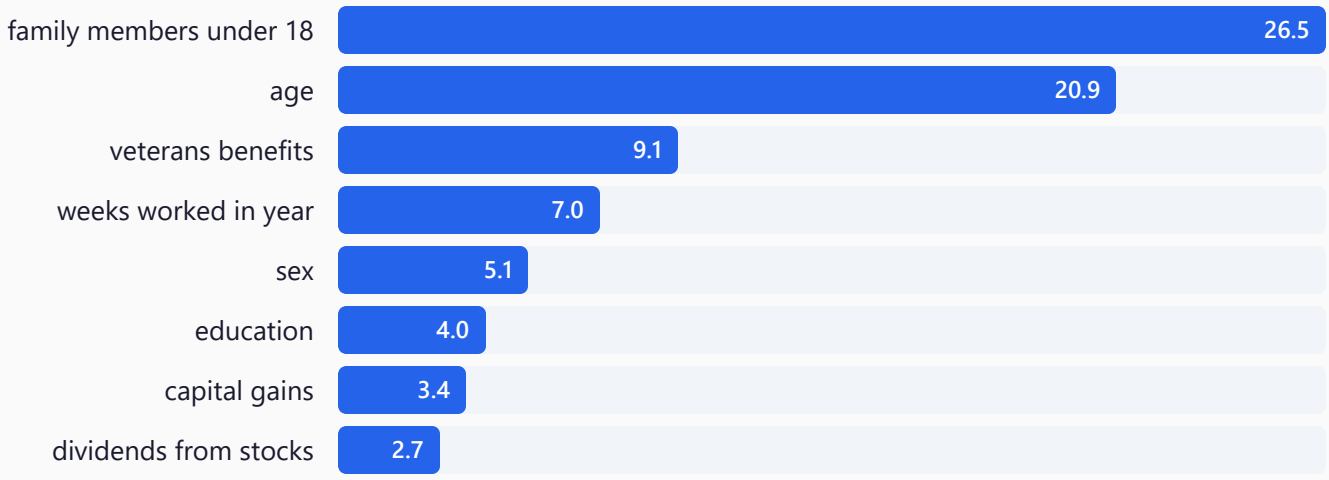
Selected: depth=6, lr=0.08, l2_leaf_reg=1.0 (+0.0055 PR-AUC over Stage 5 baseline). Results consistent under survey-weighted evaluation.

Threshold Policy

The default 0.5 threshold wastes most of the model's value on this imbalanced task. Three operating modes give the business explicit control:

OPERATING MODE	THRESHOLD	PRECISION	RECALL	F1	USE CASE
Precision-leaning	0.45	0.726	0.510	0.599	Expensive 1:1 outreach
Balanced	**0.25**	**0.585**	**0.683**	**0.630**	**General targeting**
Recall-leaning	0.20	0.532	0.732	0.616	Broad screening

Feature Importance



The model relies on a diverse mix of human-capital, employment, and household signals — consistent with economic intuition. No suspicious single-feature dominance.

Calibration

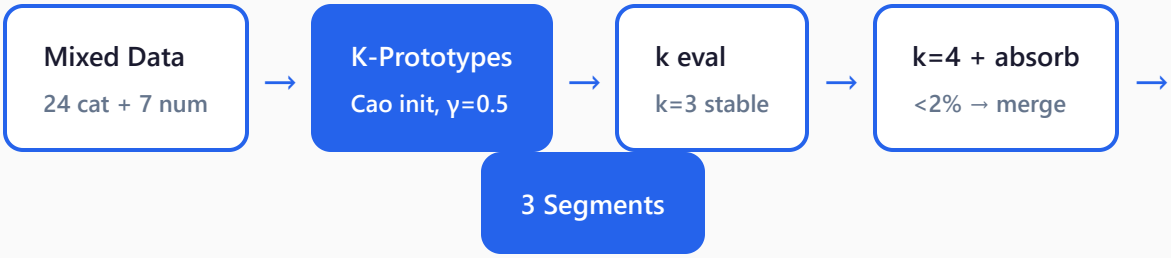
Not adopted. Brier score is already 0.031 (excellent). Neither isotonic (0.032) nor Platt scaling (0.034) improved it. Uncalibrated probabilities are well-calibrated across the full range.

5. Segmentation Modeling

Why Segment Beyond the Classifier?

The classifier tells you *who is likely high-income*. It does not tell you *who these people are* or *how to approach them*. Segmentation provides the action-design layer that converts a statistical score into a marketing plan.

Method Selection



Fit on a random 30,000-row subsample (seed=42), assigned to all 199,523 rows. In evaluation outputs, k=3 was stable while k≥4 produced micro-clusters. To make outlier handling explicit, the clustering step requested k=4, which isolated a tiny micro-cluster (<0.2%). A deterministic absorption rule then merged it into its nearest neighbor, producing the **final 3-segment solution**.

Governance

Target (`_target`) was **excluded** from clustering input — used only after clustering for profiling. Weight (`_weight`) was **excluded** from clustering input — used only for population-level segment sizing.

Final Segments

Segment 0: Senior/Retired Householders

Segment 1: Youth & Dependents

33% of population · 0.05% >$50K · 0.01× enrichment

20% of population · 2.2% >$50K · 0.35× enrichment

- Mean age 63, mostly married, not in workforce
- High school education dominant
- ~2 weeks/year worked

Marketing: Retirement products, estate planning, Medicare supplements. Direct mail and phone channels.

- Mean age 11, children and young dependents
- Not in labor force
- 57% with both parents present

Marketing: Family-indirect via parents/guardians. Youth savings, education products. Long-horizon relationship building.

Segment 2: Mid-Career Working Householders

47% of population · 12.5% >$50K · 2.1× enrichment

- Mean age 39, primarily full-time employed
- Majority married householders
- ~49 weeks/year worked

Marketing: Premium financial products, investment accounts, wealth management. Digital and advisor channels.

6. Integrated Targeting Strategy

The Two-Layer Model

LAYER	WHAT IT DOES	QUESTION IT ANSWERS
Classifier	Scores each person by income likelihood (0–1)	*Who should we prioritize?*
Threshold Policy	Converts score into a targeting decision	*Where do we draw the line?*
Segmentation	Groups people by life-stage and household type	*How should we approach them?*

Example Operational Workflows

Scenario 1: Premium credit card launch (budget-constrained)

Use **precision-leaning threshold (0.45)**. Contact only the highest-confidence prospects. Within that group, customize messaging by segment: investment language for Segment 2, retirement messaging for Segment 0.

Scenario 2: Financial literacy campaign (broad reach)

Use **recall-leaning threshold (0.20)**. Cast a wide net to find as many high-income individuals as possible. Segment determines channel: digital for Segment 2, community events for Segment 0, family-targeted for Segment 1.

Scenario 3: Market sizing (no outreach needed)

Use **segmentation alone**

. Report weighted population shares to size market opportunity by life-stage group. The classifier's probability score provides a continuous dimension for sub-targeting within the large working-age segment.

7. Key Insights

1. Threshold is a business policy, not a model parameter

The model's ranking quality (PR-AUC 0.70) is constant. What changes with threshold is the precision–recall tradeoff. The business should choose which error is more costly: missing a high-income person, or wasting outreach on someone who isn't.

2. Marginal cases are where the value concentrates

People far from the decision boundary are easy calls. The hard — and most business-relevant — decisions are about individuals the model scores near the threshold. A hard 0/1 label hides this nuance; a score with policy preserves it.

3. Income is a proxy target

The >$50K label approximates purchasing power but is not what a retail client ultimately optimizes for. In production, validate against response rates, conversion, or lifetime value — not just income classification accuracy.

4. Segmentation converts score into action

A probability score alone doesn't tell a campaign manager what creative to use or what product to offer. Segments provide the "who are these people" context that makes the score actionable.

5. Governance and interpretability beat complexity

The reduced feature set outperformed the full set. Calibration wasn't needed. The 3-segment solution is more useful than a 7-cluster one with micro-groups. At every stage, simpler and more auditable won.

8. Limitations & Next Steps

Limitations

LIMITATION	IMPACT
Historical data (1994–1995)	Demographics and income distributions have shifted. Model demonstrates methodology, not production readiness.
Income as proxy target	Real-world targeting should validate against business outcomes (response, conversion, LTV).
Segmentation is a lens	K-Prototypes with different parameters or features would produce different valid segments.
No external validation	All evaluation is within the CPS dataset; no holdout from a different source.
Single random seed	All experiments use seed=42. Multi-seed robustness not formally tested.

Recommended Next Steps

#	ACTION	WHY
1	Validate against real business outcomes	Replace income proxy with response/conversion data
2	Refresh with current data	1994 demographics don't reflect today's population
3	Sub-segment working-age group	47% of population; classifier score gives continuous sub-targeting
4	Build a response/propensity model	More directly actionable than income classification

5	Operationalize threshold selection	Give campaign managers a tool to choose their operating point

References

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Census Income Classification & Marketing Segmentation — ML Take-Home Project